

FUNDAMENTUM TRACK · HACKATHON RESEARCH DOSSIER

Foundational Problems at Quantitative Trading Firms

—

A full research sweep of the problems quant trading firms actually live with, built entirely from public primary sources: firm-published engineering material, regulatory orders and court filings, peer-reviewed papers, and industry reports. With ranked project recommendations and the complete citation trail.

Prepared: 21 August 2026

Sources: 130+ cited items across 7 research areas, key items fetched and verified

Structure: Part 1 is the skim layer. Parts 2 and 3 are the comprehensive layer. Part 4 is the citation trail.

Orientation

This dossier has a skim layer and a deep layer. If you have ten minutes, read Part 1 only; it contains the whole argument. When you have picked a direction, read that direction's section in Part 2 and its project write-up in Part 3. Part 4 is the full numbered source list, which doubles as your research trail for submission.

Part 1 · The Skim Layer (extended TLDR) · the problem map, top project picks, what the track brief claimed vs what the evidence showed, and how to win the on-theme prize

Part 2 · Problem Statements in Detail · seven problem areas, each with the evidence, the numbers, the existing tools and their gaps, and a buildability verdict

Part 3 · Project Recommendations · four ranked primary recommendations with build plans and demo scripts, plus five alternates

Part 4 · Research Citations · the complete source list, coded and grouped, with key claims and URLs

Appendix · Standing Context · the assignment, constraints, glossary, and next steps, so this document works standalone

Provenance marking

Anything in this document that originates from the hackathon track brief itself (the case study you were handed) is marked **BRIEF**. Marked material is the organisers' framing, not evidence, and should never be cited in your submission as research. Everything unmarked carries a citation code like [R7] or [F13] pointing to a numbered source in Part 4. Citation code prefixes: **R** reproducibility and backtest overfitting, **A** alpha decay and model drift, **D** market data quality, **P** plumbing and practitioner sweep, **O** operational and regulatory, **T** talent and knowledge transfer, **F** firm-published technical material.

Method, honestly stated

This research was done without interviews. The substitute is public primary material, weighted in this order: what elite firms publish about their own engineering problems (blogs, papers, conference talks, podcast transcripts from Jane Street, Optiver, IMC, Hudson River Trading, Two Sigma, Man Group, G-Research, XTX, WorldQuant, Tower Research); what regulators and courts have documented (SEC, CFTC, FCA, FINRA, ESMA orders and reviews, court filings); peer-reviewed and working papers with hard numbers; and industry reports. Forum material (Hacker News, QuantNet, Blind) is used sparingly and only as colour, never as a load-bearing source. Note on Millennium: it publishes essentially no technical material publicly; its affiliate WorldQuant is the closest primary channel and is cited where relevant [F25, F26, T13].

The Skim Layer

1.1 The one-paragraph conclusion

The most winnable problems for this track share three properties: a firm or regulator has admitted the problem in writing, the evidence contains hard numbers you can put on a slide, and a working prototype fits in a weekend on free data. Four problems clear that bar decisively: **signal decay monitoring** (a landmark paper gives you a labelled benchmark to replay), **backtest integrity** (the SEC has fined firms for exactly the bug you would detect), **market data quality under stress** (official incident postmortems give you real disaster days to replay), and **deterministic testing of trading systems** (the pain three elite firms confess to most, with the emptiest tooling landscape). The talent and knowledge-transfer direction, which the track brief recommends most strongly **BRIEF**, is still viable without interviews because firms litigate over departing knowledge, and court filings beat coffee chats; but its inside-the-building texture is the one thing public sources cannot fully replace.

1.2 The problem map

Every problem found in the sweep, rated on three axes. Evidence strength asks: is there primary, quotable, numbered proof this problem is real? Buildability asks: can two to four strong CS students demo a credible prototype in a weekend on free data? Competition asks: how likely are other teams in the room to pick it (lower is better)?

PROBLEM	ONE-LINE STATEMENT	EVIDENCE	BUILDABLE	COMPETITION
Backtest overfitting and reproducibility	Ordinary research workflows manufacture false discoveries, and nobody records the number of trials that would let you correct for it [R1, R3, R7]	Strong	High	Medium
Signal decay and crowding	Published signals lose over half their returns post-publication, and the industry's actual decay detector is losing money first [A1, A15, A20]	Strong	High	Low-Med
Market data quality under stress	Feeds degrade exactly when volatility spikes; even exchanges have gone 90 minutes without noticing their own failures [D1, D5, D6]	Strong	High	Medium
Testing and simulation of trading systems	Non-deterministic, market-facing systems cannot be validated by unit tests; every firm hand-rolls simulators and there is no off-the-shelf exchange-in-a-box [F8, F9, F11]	Strong (firm-confessed)	Medium	Very low
Silently failing alerts	Alert systems rot at scale and fail silently when metrics disappear; Knight Capital ignored 97 error emails on its way to a \$460M loss [P5, O1]	Strong	High	Very low

Algo governance and change control	Regulators found firms cannot even inventory their own algorithms; one untracked model change cost Two Sigma \$255M in penalties and repayments [O2, O10, O11]	Strong	High	Very low
Knowledge transfer and talent churn	Strategies live in heads; two departures preceded a >50% profit drop in one month on a billion-dollar strategy, per a federal court filing [T1, T2, T8]	Strong (headline), weaker (texture)	Medium	Medium (brief steers teams here)
Pre-trade risk controls	The FCA spelled out how a \$444B fat-finger basket got past click-through alerts; the required control stack is public rule text [O3, O14]	Strong	High	Low-Med
Symbology and instrument mapping	The same instrument has different identifiers per venue and vendor, some remapped daily; every firm re-solves this [P8, P9]	Medium	High	Very low
Exchange calendars and schedules	The whole industry hand-patches holidays and session changes forever; the proof is a public changelog of endless corrections [P11]	Medium	High	Very low
Corporate actions	Only 4 in 10 voluntary corporate actions achieve straight-through processing; vendors silently restate adjusted price history [P15, O18]	Medium	High	Very low
Regulatory reporting breaks	Citadel Securities misreported 42 billion order events across 33 error types; reporting accuracy defeats even the best firms [O6]	Strong	Medium	Very low
Research data infrastructure	Six elite firms each built bespoke petabyte-scale research data platforms because nothing off the shelf fits [F18-F24]	Strong	Low (solved-ish, huge scope)	Low
Market data licensing and entitlements	Per-exchange fees, per-user reporting and audit exposure; a pure administrative tax every firm quietly staffs [P16, D11]	Medium	Medium	Very low

*Deliberately excluded, as the track brief also advises **BRIEF**: raw latency (years and millions in hardware; unresearchable IP) and novel alpha signals (what firms hunt full-time; also unresearchable). Our sweep independently confirmed the unresearchability: the closest firms come to discussing either in public is stylised conference talks [F2, F3].*

1.3 Top project picks

Pick 1 • Signal Decay Early-Warning System. Replay a landmark result live: the Chen-Zimmermann open dataset gives monthly returns for 205 published anomalies with each paper's sample-end and publication dates [A26], so you can run changepoint detectors over each signal as if in real time and measure, in months, how much earlier a proper monitor flags decay than the industry default of noticing the P&L. McLean and Pontiff's 26% out-of-sample and 58% post-publication decay numbers [A1] are your problem slide; Man Group's statement that slow decay is harder to detect than crashes [A15] is your motivation quote.

Uniquely for a monitoring project, you have a labelled benchmark, so your demo can end with a leaderboard of losses avoided. Full write-up: Part 3, Project 1.

Pick 2 • Backtest Integrity Toolkit. A lookahead-bias and survivorship taint-checker plus an ambient trial ledger that computes the Deflated Sharpe Ratio automatically. The SEC fined F-Squared \$35M over a backtest inflated by exactly the lookahead bug your linter catches [R28]; Quantopian's own study of 888 real algorithms found in-sample Sharpe predicts out-of-sample performance with R-squared under 0.025, and that the more backtests a user ran, the worse the gap [R7]. The deflation formulas exist in open source but are integrated into no backtester [R26]; that missing integration is your product. Full write-up: Part 3, Project 2.

Pick 3 • Feed Health Monitor. A streaming scorecard of market-data integrity (sequence gaps, stale and crossed quotes, price-band violations, event-time vs receive-time skew) demoed against a replay of a real disaster day, such as 3 June 2024, when a SIP software update showed Berkshire Hathaway down 99.97% [D6]. The official flash-crash report documents that data-integrity doubt itself withdrew liquidity [D1], and the SEC's 2026 order against NYSE shows the exchange itself went about 90 minutes without detecting its own auction failure [D5]. Free replay data exists (IEX pcaps, LOBSTER samples, Binance depth, Databento trial credit) [D22-D26]. Full write-up: Part 3, Project 3.

Pick 4 • Deterministic Market-Replay Test Harness. The sleeper with the strongest firm-primary evidence: Jane Street, HRT and Optiver all publish about how hard it is to test non-deterministic trading systems [F8, F9, F11, F12], Jane Street devoted a podcast episode to a deterministic-testing startup [F9], and our sweep found no off-the-shelf exchange-in-a-box test harness. Knight Capital is the canonical cost of the gap: dead code, untested redeploy, \$460M in 45 minutes [O1]. Highest novelty, near-zero competition, moderately harder build. Full write-up: Part 3, Project 4.

Sleeper alternates (Part 3.5, shorter write-ups): an alert linter that catches silently-failing alert rules [P5, O1]; an algo register and change-audit tool that drafts the RTS 6 self-assessment regulators call burdensome [O10-O12]; an exchange-calendar consistency engine [P11]; a corporate-actions diff engine that detects silently restated price history [P15, O18]; and a knowledge bus-factor dashboard built on the litigation record [T1, T8].

1.4 Testing the brief's own claims

The track brief makes factual claims about the industry **BRIEF**. Since the on-theme prize rewards showing how your starting assumption changed, we treated the brief itself as the starting assumption and checked it against primary sources. This table is submission-ready evidence of that process.

CLAIM IN THE BRIEF **BRIEF**

VERDICT

WHAT THE INDEPENDENT EVIDENCE SAYS

"A strategy that shone in someone's notebook can't be rebuilt months later because the data, parameters, and code weren't captured"	Confirmed, strengthened	Microsoft's flagship quant-ML framework cannot reproduce its own published benchmarks (open GitHub issues) [R23]; Man Group built an entire database (ArcticDB) to reconstruct what data looked like at research time [R14, F20]; the SEC's Two Sigma order shows unauthorized changes to 14+ live models going undetected for two years [O2].
"Most academic trading strategies don't survive contact with real capital, largely from overfitting and lookahead bias"	Confirmed with numbers	26% out-of-sample and 58% post-publication decay across 97 predictors [A1]; 65% of 452 anomalies fail replication under stricter tests [A3]; a practitioner who re-implemented 130 papers reported essentially all were p-hacked or overfit [R12]; and the SEC has fined a firm \$35M over a lookahead-inflated backtest [R28].
"Market-data systems that are fine on a calm day fall over in high volatility, exactly when they matter most"	Confirmed with incidents	Nasdaq's SIP collapsed under quote bursts 26x its tested capacity [D3]; options feed microbursts now exceed 70 Gbps, beyond what most firms can receive without gaps [D12]; the VIX itself spiked 180% partly as a quote-quality artifact [D13].
"When a quant leaves, their undocumented knowledge often leaves with them" and firms "run on high churn"	Confirmed (headline), partially (texture)	Federal court filing: a strategy built by ~25 people saw profits drop >50% the month after two departures [T1]; 15-20% annual PM turnover and 55% three-year retention at multi-managers [T8, T9]; but the inside-view texture (onboarding pain, dead wikis) exists publicly only in low-citability forums, which is the one gap interviews would have filled.
"Knowing a strategy is decaying before it costs you" is a monitoring problem, not a modelling problem	Confirmed, and the gap is real	Man Group: slow alpha decay is harder to detect than acute crashes [A15]; practitioner kill-criteria exist but are contested math [A8, A22]; no mainstream ML monitoring product (Evidently, NannyML, Arize) ships anything trading-aware [A24].

1.5 How to win the on-theme prize with this dossier

The brief says the prize goes to the clearest line from foundational research to final build **BRIEF**. Concretely, with this material:

- **Keep a decision log.** For every build decision, write one line: "We did X because source Y says Z." Part 3 lists these evidence chains explicitly for each recommended project; lift them.
- **Show the assumption that died.** Use section 1.4, or better, add your own: run one of the brief's claims into a primary source and note where the source surprised you.
- **Cite primaries, not paraphrases.** The brief warns against reading summaries **BRIEF**. Part 4 gives you direct links to SEC orders, court filings, firm engineering posts, and papers. Quote those, with paragraph or page references where we noted them.
- **Name the existing tools and where they fall short.** Each Part 2 section has a tools-and-gaps subsection with citable evidence of the gap, which is the part most teams skip.
- **Let a firm say your problem statement for you.** The strongest rhetorical move available: open the pitch with an elite firm's own words (for example HRT admitting its alerts "rarely handled timeseries absence correctly" [P5], or Jane Street on tests missing "the critical scenario that will expose your mistake" [F8]).

1.6 Caveats and integrity notes

- Numbers we could not verify against the fetched full text are flagged in Part 2 and Part 4; double-check anything you put on a slide against the linked primary.
- One recent paper on alpha decay was withdrawn by its author for revision [A11]; cite it only as evidence the problem is open, never for its numbers.
- Some outlets (Bloomberg, AQR PDFs, Wiley, Taylor & Francis) block automated fetching but open fine in a normal browser; those are marked in Part 4 so you can pull them manually.
- Forum threads are included in the citation list for completeness but are marked as colour-grade sources; your submission should lean on the firm-published and regulatory material, which is both stronger and closer to how the practitioners you cannot interview actually communicate.
- This dossier itself is a research aid. For the submission, present the primary sources in Part 4 as your trail, plus your own reading notes and decision log on top of them.

Problem Statements in Detail

Seven problem areas. For each: where the direction came from, the problem stated precisely, the evidence with numbers, the existing tools and where they fall short, and a verdict on hackathon fit. Citation codes resolve in Part 4.

2.1 The reproducibility gap and backtest overfitting

Origin: suggested as a direction in the track brief **BRIEF**. Everything below it is independent evidence from our sweep.

The problem

A quantitative research process that tries many strategy configurations and reports the best one will systematically manufacture false discoveries, because the standard performance statistics assume one trial. Bailey, Borwein, Lopez de Prado and Zhu proved the size of the effect: try just 10 configurations of a strategy with zero true skill and you should expect to find one with an in-sample Sharpe ratio of 1.57; with five years of data, trying more than about 45 independent configurations near-guarantees a backtest Sharpe of 1 with expected out-of-sample Sharpe of zero [R1]. The correction exists (the Deflated Sharpe Ratio deflates reported performance by the number of trials [R3]), but it requires knowing N, the number of things tried, and no standard research workflow records N. The second face of the same problem is reconstruction: a result produced in a notebook months ago cannot be rebuilt because the data as it looked then, the parameters, and the code state were never captured together.

Why it persists

Because the missing ingredient is a process discipline, not an algorithm. Lopez de Prado's widely-read account of why machine-learning funds fail names the "Sisyphus paradigm": siloed researchers each running thousands of undocumented backtests, with selection bias unrecorded at the firm level [R6]. Harvey, Liu and Zhu catalogued 316 published factors and concluded, in the paper's own words, that "most claimed research findings in financial economics are likely false" [R4]; the president of the American Finance Association made the same argument to the profession directly [R5]. A checklist protocol exists (Arnott, Harvey and Markowitz: log every test, fix hypotheses ex ante, track data lineage [R9]) but there is no tooling that enforces it ambiently.

The evidence

KEY NUMBERS

- 888 real user algorithms on Quantopian: in-sample Sharpe predicted out-of-sample performance with R-squared below 0.025, and the more backtests a user ran, the larger the in-sample vs out-of-sample gap [R7].
- SEC v. F-Squared: a \$35M settlement over a marketed track record that was a backtest, itself miscalculated by applying signals to prices before the signals could have existed. Classic lookahead bias; the firm went bankrupt [R28].
- SEC v. AXA Rosenberg: \$242M over a concealed coding error in a quant risk model that caused \$217M in client losses [R27].
- Survivorship bias demo from fund principal Ernest Chan: the same strategy returns +388% on survivorship-biased data and -42% on the true historical universe [R16].
- Index products marketed on backtests: 12.25% average annual excess return in the five years before index launch, negative 0.26% in the five years after (Vanguard study); a Morningstar study of 782 indexes found the same pattern [R30].
- A practitioner who re-implemented 130 academic prediction papers: "Literally every single paper was either p-hacked, overfit, or a subsample of favourable data was selected" [R12].

On the reconstruction face: Microsoft's qlib, the highest-profile open quant-ML framework, has multiple open issues stating its own published benchmarks cannot be reproduced [R23]. Man Group built ArcticDB, a versioned bitemporal DataFrame database, explicitly because reproducing research requires reconstructing exactly what the data looked like at research time [R14, F20]. When Quantopian shut down, the dominant open-source backtester (zipline) was orphaned and strategies written against it could not be rerun until volunteers forked it [R15, R22].

Existing tools and the gap

Generic experiment trackers (MLflow, Weights & Biases) and data versioning tools (DVC, lakeFS) exist but are built for ML-in-production teams: they are heavy for notebook-scale research, and none of them track the quant-specific unit of record: data as-of date, universe definition, trial count against a hypothesis [R24, R25]. The deflation statistics (PBO, DSR) exist as standalone open-source packages [R26] but are integrated into no backtester and no tracker. The gap, stated plainly: nothing in the open ecosystem counts your trials and deflates your Sharpe for you.

Verdict

Evidence: strong (regulatory cases, platform-scale data, canonical papers).

Buildability: high. Competition: medium (backtesting-adjacent projects are common, but an integrity tool inverts the usual toy; the brief itself warns that a generic backtester is a weak entry **BRIEF**). See Part 3, Project 2.

2.2 Alpha decay, crowding, and model drift

Origin: suggested as a direction in the track brief **BRIEF**. Everything below it is independent evidence from our sweep.

The problem

Trading signals are a depleting resource: they decay as they are published, crowded, and arbitrated, and they break abruptly when market regimes change. The operational problem for a firm is not building a better signal but knowing that a running strategy is dying before the P&L makes it obvious. Man Group's research group states the crux: crowding produces slow alpha decay plus acute left-tail liquidation risk, and "left tail events can be very acute in a very short space of time, while alpha decay is much slower, making it harder to detect" [A15].

The evidence

KEY NUMBERS

- Across 97 published predictors: returns 26% lower out-of-sample and 58% lower post-publication [A1].
- The August 2007 quant quake: a canonical contrarian strategy's daily return eroded roughly 90% between 1995 and 2007 as competition crowded in, forcing leverage up; an 8:1-levered version would have lost about 31% in two days during the coordinated unwind [A2].
- Momentum crashes: in 2009, the losers leg returned +163% against +8% for winners over three months; regime-conditioning the same signal roughly doubles its Sharpe, proof that monitoring and conditioning have real value [A5].
- Replication war: 65% of 452 anomalies fail under stricter tests [A3]; the strongest rebuttal still finds roughly a 47% post-publication decline in alpha, just not to zero [A4].
- The quant winter: only about 15% of quant funds beat benchmarks in 2018-2019 [A16]; a major factor fund shrank 92% from its 2018 peak [A18]; in 2020 Renaissance's internal Medallion fund returned +76% while its outside funds lost 20-34%, with the firm citing being wrongly hedged through the regime change [A17].
- A prop firm (Maven Securities) published its own decay measurements: signal decay costs averaging 9.9% per year in Europe and 5.6% in the US, and rising [A20].

How decay is detected in practice is the revealing part. The public practitioner material on kill-criteria is thin and contested: a CFM paper derives when a drawdown is statistically too deep for the claimed Sharpe [A8], a practitioner tool computes the p-value of the current drawdown against the backtest [A21], and a practitioner critique notes even these rest on assumptions real P&L violates [A22]. The honest summary of the status quo: the industry's default decay detector is losing money.

Existing tools and the gap

ML model-monitoring products (Evidently, NannyML, Arize, WhyLabs) monitor input-distribution drift and prediction accuracy against delayed labels. A trading strategy fails differently: inputs can be perfectly stationary while the relationship between signal and

forward returns quietly dies from crowding. The failure signature is information-coefficient decay, rising co-movement among strategy holdings (the measurable crowding proxy of Lou and Polk [A7]), and regime breaks; none of that maps to "data drift," and NannyML's own material contains no finance use case at all [A24]. The methods needed are weekend-scale: CUSUM and Bayesian online changepoint detection are each implementable in about a hundred lines [A9]. The data needed is free and, uniquely, labelled: the Chen-Zimmermann Open Source Asset Pricing dataset ships monthly returns for 205 published anomalies with sample-end and publication dates [A26], plus the Ken French library and the JKP global factor data [A25, A27].

Verdict

Evidence: strong. Buildability: high, with a labelled benchmark almost no monitoring project ever has. Competition: low to medium. See Part 3, Project 1.

2.3 Market data quality and infrastructure under stress

Origin: suggested as a direction in the track brief **BRIEF**. Everything below it is independent evidence from our sweep.

The problem

Market data systems degrade precisely when they matter most. Volatility spikes multiply message rates past tested capacity, feeds drop packets and desynchronize, bad prints and stale quotes propagate into every consuming system, and, in the official record of the 2010 flash crash, doubt about data integrity itself caused firms to pull liquidity, amplifying the crash [D1]. By design, exchange feeds are UDP multicast and gap-handling is the subscriber's problem: Jane Street's own podcast explains that exchanges "pile forward obliviously... if people are behind, they need to engineer their systems differently to keep up" [D17].

The evidence

KEY NUMBERS AND INCIDENTS

- Flash crash, 6 May 2010: over 20,000 trades busted as clearly erroneous, executions at \$0.01 and \$100,000, and documented delays in consolidated quote data; firms' data-integrity checks triggered trading pauses [D1].
- Nasdaq "flash freeze," Aug 2013: quote bursts exceeding 26,000 updates per port per second against a tested capacity of 10,000 collapsed the SIP and halted all Nasdaq-listed trading for about three hours; the failover also failed [D3].
- NYSE, 24 Jan 2023: opening auctions failed in over 2,800 securities; the SEC's 2026 order records that NYSE was unaware of the full scope for roughly 90 minutes and paid a \$9M penalty for lacking monitoring policies [D5].
- 3 June 2024: a software update at the consolidated tape's limit-up-limit-down system displayed Berkshire Hathaway down 99.97% and triggered erroneous halts across dozens of symbols [D6].
- 28 Nov 2025: a data-center cooling failure halted essentially all CME futures trading for about 10 hours [D7].
- Options feed (OPRA) volumes after the Feb 2024 expansion: 130-145 billion messages per day, with one-millisecond microbursts exceeding 70 Gbps, beyond what most subscribers can receive without gaps [D12].
- The August 2024 VIX spike of 180% was, per BIS analysis, substantially a quote-quality artifact: the index is computed from quotes, and dealers' pre-market spread-widening mechanically inflated it [D13].
- The SEC itself formally found the consolidated feeds slower and thinner than proprietary feeds, a two-tier market codified in its 2020 Market Data Infrastructure rule [D8]; academic measurements quantify the dislocations [D9, D10]. Industry spend on market data reached \$42B in 2023 and kept rising [D11].

Two vendor-published pieces sharpen the problem usefully. Databento (founded by ex-HFT engineers) argues you should not silently "clean" market data at all, because naive cleaning desynchronizes backtest from live; bad data concentrates in redistributor and vendor feeds, and the right posture is detection and flagging [D14]. The same firm documents that vendor daily volumes routinely fail to reconcile with tick-level sums, and settlement-price timing quietly injects lookahead bias into daily datasets [D16].

Existing tools and the gap

General-purpose data validation frameworks (Great Expectations, pandera, deequ) are batch, schema-and-statistics tools with no concept of exchange sequence numbers, A/B feed arbitration, crossed books, price bands, or event-time vs receive-time skew [D21]. Market-data-native quality tooling exists only inside closed commercial products or firms' in-house code [D18]. There is no open-source streaming tick-data sanity layer.

Verdict

Evidence: strong (official postmortems with numbers). Buildability: high, with dramatic replay demos on free data. Competition: medium. See Part 3, Project 3.

2.4 Testing and simulation of trading systems

Origin: NOT in the track brief. This direction emerged from mining what elite firms publish about their own engineering, which was the strongest multi-firm confessed pain with the emptiest tooling landscape.

The problem

Trading systems are non-deterministic, stateful, market-facing programs whose worst failures occur in scenarios no unit test anticipates, and which can never be stopped for testing. Jane Street's engineering blog states it directly: "It's all too easy for your tests to miss the critical scenario that will expose your mistake" [F8], and its podcast episode on testing lays out why: with threads and sockets "suddenly your computer program is not a pure function... It's a big, complicated, weird state machine" [F9]. HRT publishes on having to simulate and verify FPGA trading logic before it touches the market [F11]; Optiver publishes on the pain of building CI observability for trading code [F12]; Jane Street separately publishes on modernizing a mission-critical system that can never go down, "swapping the engine out of a moving race car" [F30].

The evidence

KEY FACTS

- Knight Capital, 1 Aug 2012: a deployment left dead code active on one of eight servers; a repurposed flag triggered it; 212 parent orders became about 4 million executions and a loss above \$460M in roughly 45 minutes. The SEC order itemizes the missing controls and notes ignored pre-open error emails [O1]. This is what an untested deployment path costs.
- Three elite firms publish about the testing problem in their own voices [F8, F9, F11, F12], and a fourth data point: Jane Street hosting the founder of Antithesis (deterministic-hypervisor testing) on its own podcast is itself evidence the firm considers the problem unsolved [F9].
- Regulation now demands what tooling does not supply: MiFID II RTS 6 requires firms to test algorithms in non-live environments and document it, and supervisory reviews found firms could not adequately describe their test environments [O11, O12].

Existing tools and the gap

Every firm hand-rolls its own simulator; there is no off-the-shelf deterministic exchange-in-a-box that replays real market data, injects faults (gaps, duplicate packets, out-of-order delivery, exchange outages mid-session), and produces reproducible runs. The only generic entrant, Antithesis, is domain-agnostic and priced for enterprises. Open backtesters simulate strategies against clean bars; they do not simulate the market infrastructure misbehaving.

Verdict

Evidence: strong and unusually pure (firm-authored confessions plus a \$460M regulatory case study). Buildability: medium (scope control is the risk).

Competition: near zero. See Part 3, Project 4.

2.5 The unglamorous plumbing layer

Origin: NOT in the track brief. Found by an open-ended sweep for foundational problems practitioners and firms describe that no one puts on recruiting posters.

The problem, as a cluster

Beneath strategies sits a layer of permanent, structural toil that every firm staffs and nobody celebrates. Five members of the cluster stand out, each independently buildable:

a. Alerts that fail silently

HRT, which runs thousands of alerts across dozens of teams, published that its alert definitions suffered "excessive code duplication, poor expressiveness... and subtle footguns," and specifically that alerts "rarely handled timeseries absence correctly": when a metric disappears, the alert that watches it silently stops firing [P5]. The catastrophic version is in the SEC record: Knight Capital's systems sent 97 automated error emails before the market opened on the day it lost \$460M; they were not acted on [O1]. Monitoring at trading scale also breaks the standard stack outright [P6, F14, F15].

b. Symbolology and instrument identity

The same instrument carries different identifiers per venue, per vendor, and per day; Databento's engineering guide notes some publishers remap numeric instrument IDs daily [P8]. Bloomberg itself has campaigned publicly against the licensing-encumbered identifier regime (CUSIP fees) [P9]. Every backtest bug involving a renamed or reused ticker traces here, and builders keep independently re-solving the mapping problem [D19].

c. Exchange calendars and schedules

The proof is elegant: the public changelog of pandas-market-calendars, the standard open-source library, is an unending stream of hand-patched corrections (revised SIFMA holidays, changed JPX close times, a coronation holiday) [P11]. The world's exchanges never stop changing their schedules, and the industry's maintenance mechanism is a volunteer noticing.

d. Corporate actions

DTCC processes over 600,000 voluntary reorganization instructions a year and states that firms maintain "large teams of personnel" specifically to monitor them for costly errors; only about 4 in 10 voluntary corporate actions achieve straight-through processing [P15, O18]. Downstream, vendors silently restate split- and dividend-adjusted price histories, which invalidates backtests that consumed the earlier version [P15].

e. Legacy systems, vendor lock-in, and the rest

Also documented in firm and practitioner sources: mission-critical decades-old systems that can never be paused for replacement [P1, F30]; a whole industry stuck on kdb+/q, with roughly ten firms exploring migrations and none pulling the trigger [P13]; Optiver spending 30-40% of engineering headcount consolidating internal platforms [P3]; Excel as load-bearing infrastructure at sophisticated firms [P17]; market data licensing and entitlement

reporting as an audit-carrying administrative tax [P16]; and clock synchronization as both a debugging and a regulatory problem, with MiFID II requiring demonstrable sync within 100 microseconds of UTC [P12, F28].

Verdict

Evidence: medium to strong depending on the member. Buildability: high for alerts, calendars, symbology, corporate actions. Competition: near zero for all of them. These are the picks if your team's priority is being the only team in the room on your problem. See Part 3.5.

2.6 Operational, risk, and regulatory controls

Origin: NOT in the track brief. Found by mining enforcement actions and supervisory reviews, which are the most citable problem statements in the whole domain: a regulator documenting a failure is a problem statement with a dollar figure attached.

The problem

The controls layer (pre-trade risk checks, algo inventories and change control, surveillance, regulatory reporting) is where sophisticated firms fail expensively and repeatedly, and the public record is extraordinarily specific about how.

The evidence

THE ENFORCEMENT RECORD

- **Knight Capital (SEC, 2013):** first Market Access Rule enforcement; the order itemizes control-by-control what was missing: no point-of-entry erroneous-order checks, no firm-wide capital thresholds, deficient review and certification [O1].
- **Two Sigma (SEC, 2025):** employees flagged model-parameter permissioning vulnerabilities in March 2019; not remediated until August 2023. One employee made unauthorized changes to 14+ live models. \$90M penalty plus \$165M voluntarily repaid [O2].
- **Citigroup (FCA/PRA, 2024):** a trader meant to sell a \$58M basket and created a \$444B one; \$189B reached the trading algorithm and about \$1.4B executed, briefly crashing European indices. The regulators' named failures are a design spec: no hard block, alerts dismissable without reading, monitoring too slow [O3].
- **CAT reporting (FINRA):** Citadel Securities misreported about 42.2 billion order events across 33 distinct error types; IMC about 21.8 billion across 35 types, including 680M+ orders missing millisecond precision [O6].
- **Latour Trading (SEC, 2015):** a single coding change silently broke order-compliance logic for years; roughly \$8M in fines [O7].
- **Tower Research (CFTC/DOJ, 2019):** \$67.4M spoofing settlement, then a record for a prop firm; surveillance of your own traders is a prop-shop problem too [O4].

The supervisory reviews are as valuable as the fines. The FCA's 2018 multi-firm review found firms "lacked a suitable process to identify algorithmic trading across their business" (no complete algo inventory) and kept "limited documentation" of development,

testing and deployment [O10]. The Central Bank of Ireland found RTS 6 self-assessments deficient and, memorably, that firms had no formalized plan for what happens after the kill switch is pulled [O11]. ESMA itself concedes the annual self-assessment is "burdensome, in particular for smaller firms" [O12]. Surveys of prop-firm executives quantify the burden: 92% find reporting increasingly burdensome, and surveillance analysts spend 30+ hours a week manually closing alerts [O20, O21].

Existing tools and the gap

Reg-tech exists but is enterprise-priced and consultant-shaped (surveillance platforms, annual-review consultancies) [O22]. Nothing developer-native exists at small-firm price for the specific, well-specified jobs the reviews describe: maintaining an algo inventory from the repo outward, logging change control, drafting the RTS 6 self-assessment, or replaying famous disasters against a pre-trade risk gateway to score it clause-by-clause against SEC Rule 15c3-5 [O14].

Verdict

Evidence: the strongest citability in this dossier (orders with paragraph numbers and dollar figures). Buildability: high (synthetic data suffices; no market data needed). Competition: near zero. See Part 3.5 (Algo Register, Break Triage) and Part 3 Project 4 (the replay harness overlaps here).

2.7 Talent churn and knowledge transfer

Origin: the track brief's flagship suggestion, recommended there because teams could interview peers and alumni **BRIEF**. Your team cannot interview, so this section reassesses the direction on public evidence alone.

The problem

Quant firms run on high churn, and the knowledge that generates returns lives substantially in people rather than documents, so departures move P&L. The strongest single piece of evidence in this entire dossier is a federal court filing: Jane Street's complaint against Millennium describes a strategy built by about 25 people over years, "incredibly valuable" and IP-sensitive, whose profits dropped more than 50% in the month after two traders departed; the legal fight then turned on whether the secret sauce was the firm's trade secret or the traders' own "experience and expertise," which is the tacit-knowledge problem stated under oath [T1]. The strategy at issue generated about \$1B in 2023 [T2].

The evidence

KEY NUMBERS

- Multi-manager churn: roughly 15-20% annual PM turnover; three-year retention around 55%; Millennium described as cutting PMs at 7.5% drawdowns [T8, T9].
- Non-competes as knowledge dams: Citadel now imposes one to two years even on some analysts (Bloomberg, Aug 2026) [T4]; firms pay five figures monthly to enforce garden leave [T5]; the hedge fund industry's own lobby argued to the FTC that non-competes are essential precisely because strategies leave with people [T12].
- Historical rhyme: Millennium paid Renaissance \$20M in 2007 to settle a trade-secrets case over two hired quants [T6].
- Model opacity as the flip side: the SEC's Two Sigma case shows one researcher altering 14+ live models undetected for two years [O2, T24].
- Positive proof that structured capture works: Marshall Wace's TOPS alpha-capture system turned structured idea submission into roughly \$30B of AUM [T14]; WorldQuant industrialized alpha research into a platform where submissions are machine-legible by construction [T13, F26].

Where firms do publish, the confession is oblique but real: Jane Street devotes podcast episodes to building tools for expert users and to why internal documentation never keeps up [F34, F35, F36], and the sociological literature (based on 300+ insider interviews someone else already did) documents that HFT knowledge is tacit and socially embedded [T16].

The honest weakness

The headline layer (churn rates, lawsuits, dollar figures) is better documented publicly than what three coffee-chat interviews would yield, and court filings are adversarially-tested testimony. What public sources cannot give you is the inside texture: what onboarding actually feels like, which wikis are graveyards. That texture exists only in anonymous forums of low citability. If you pick this direction, build the problem statement entirely on the litigation and enforcement record and treat forum material as illustration. Expect other teams on this direction, since the brief steers everyone here **BRIEF**; your differentiation would be exactly the primary-source rigor above.

Verdict

Evidence: strong at the headline layer, weak at the texture layer. Buildability: medium (demos need credible synthetic corpora). Competition: medium to high. See Part 3.5 (Bus-Factor Dashboard).

2.8 What the firms themselves publish: cross-cutting summary

Origin: NOT in the track brief. Direct mining of technical publications by Jane Street, Optiver, IMC, HRT, Two Sigma, Man Group, G-Research, XTX, WorldQuant and Tower Research, per your instruction to go to the firms' own material.

Ranking the themes by how many elite firms publish about the same pain:

1. **Research data infrastructure at scale (6+ firms).** HRT built a distributed filesystem for hundreds of petabytes [F18]; Two Sigma treats data as code because dataset onboarding bottlenecks research [F19]; Man Group built ArcticDB [F20]; Jane Street built a columnar warehouse from scratch [F21]; G-Research built and open-sourced a multi-cluster batch scheduler because research compute broke Kubernetes [F22]; XTX discloses 25,000+ GPUs and over an exabyte of storage [F24]. The most-confessed problem, but the least hackathon-suitable: the confessions are also the solutions, and the scope is industrial.
2. **Monitoring and microsecond observability (3 firms).** HRT on alert rot and Prometheus at billions of samples [F13, F14]; Jane Street on sampling profilers being "approximately useless" at microsecond scale and on production debugging as detective work [F15, F16]; Optiver on CI visibility [F12].
3. **Testing and simulation (3 firms).** Covered in section 2.4; the most under-served by existing tools.
4. **Market data at wire speed (4 firms).** Jane Street, Optiver, IMC, HRT all publish on the same escalation: kernel bypass, cache-level surgery, then FPGAs, then tooling for the FPGAs [F1-F7].
5. **Legacy evolution without downtime (3 firms).** Jane Street's race-car engine swap and its build system outgrowing a 65M-line monorepo [F30, F32]; HRT hot-patching a vendor binary and forking CPython over import speed [F31, F33].

For your pitch, this section is a quarry of firm-voiced problem statements: whichever project you build, open with the matching confession.

Project Recommendations

Four primary recommendations, ranked, then five alternates. Every project includes its evidence chain in the form the judges are scoring: source, therefore decision. Estimated builds assume two to four capable CS students and a Friday-evening-to-Sunday hackathon.

"Half-Life": a signal decay early-warning system, benchmarked on 205 published anomalies

What it is. A monitoring engine that watches a trading signal's live performance stream and raises a statistically principled alarm when the signal is decaying or has regime-broken, before the cumulative P&L makes it obvious. Detectors: rolling information coefficient with Bayesian online changepoint detection, CUSUM on monthly returns, and a drawdown p-value test ("is this drawdown abnormal given the claimed Sharpe?"). The headline feature is the benchmark: run every detector over 205 published anomalies as if live, using each paper's sample-end and publication date as ground truth, and score detection lead time in months against the naive baseline of a rolling three-year Sharpe going negative.

Evidence chain (source, therefore decision).

- McLean and Pontiff measured 26% out-of-sample and 58% post-publication decay [A1], therefore decay is real, large, and dated, so detection lead time is measurable.
- Chen and Zimmermann publish the 205-anomaly monthly return dataset with dates [A26], therefore the benchmark is one CSV download.
- Man Group states slow decay is harder to detect than crashes [A15], and practitioner kill-criteria are contested math [A8, A21, A22], therefore the product is the detector, not another signal.
- No ML monitoring product ships trading-aware failure modes; NannyML's material contains no finance case at all [A24], therefore the tooling gap is defensible on the record.
- Adams-MacKay BOCPD is about a hundred lines with a public tutorial implementation [A9], therefore the method fits the weekend.

Build plan. Friday: ingest the Chen-Zimmermann CSV, normalize the 205 return streams, implement CUSUM. Saturday: BOCPD and the drawdown p-value test; the backtest-the-detectors harness; lead-time scoring. Sunday: dashboard (per-anomaly timeline showing sample end, publication, detector firings) and the leaderboard of months-of-losses-avoided; record demo.

Demo script. Open on the momentum crash or a famous anomaly's decay curve. Show the naive monitor firing years late. Show your detector firing months after publication. End on the aggregate leaderboard: "across 205 published signals, this monitor would have flagged decay a median of N months earlier than the industry default of watching the P&L."

Risks. Multiple-testing irony: you are running many detectors over many series, so report false-positive rates honestly (this also impresses judges). Monthly data limits drama; mitigate with the daily Ken French factors for one or two showcase signals [A25].

"Deflate": a backtest integrity toolkit (lookahead linter + trial ledger)

What it is. Two composable pieces. First, a leakage linter for pandas-based backtests: taint every DataFrame row with its as-of timestamp and fail the run when any signal at time t touches data stamped later than t ; plus static checks for the classic sins (negative shifts, joins on future timestamps, normalizing over the full sample, survivorship-biased universes). Second, an ambient trial ledger: a one-line import that hooks the backtest loop, content-hashes code, parameters and data per run, counts every trial against a hypothesis, and stamps each result with its Deflated Sharpe Ratio and probability of backtest overfitting.

Evidence chain.

- The SEC fined F-Squared \$35M over a lookahead-inflated backtest that defrauded investors and killed the firm [R28], therefore the linter's target bug has a regulatory body count.
- Quantopian's 888-algorithm study found trial count predicts the in-sample vs out-of-sample gap [R7], and the DSR correction requires exactly the trial count nobody records [R3], therefore the ledger records N ambiently instead of asking researchers to confess it.
- Chan's +388% vs -42% survivorship demo [R16], therefore the demo includes a survivorship check with a jaw-dropping before/after.
- PBO and DSR exist as orphan packages integrated into no backtester [R26], therefore integration, not invention, is the contribution.
- Arnott, Harvey and Markowitz published the protocol checklist practitioners are told to follow with no tooling [R9], therefore the report the tool emits is structured as that protocol.

Build plan. Friday: write a deliberately beautiful, deliberately broken demo notebook (lookahead join, full-sample normalization, survivorship universe, 200 silent parameter trials). Saturday: the linter (taint wrapper over a demo subset of pandas ops plus AST checks) and the ledger hook with DSR/PBO via existing packages. Sunday: the one-page integrity report; run it live on the broken notebook and watch the Sharpe deflate from 2.1 to roughly zero.

Demo script. Show the gorgeous equity curve. Run the toolkit. Watch it list three leaks with line numbers, reveal the 200 hidden trials, and reprint the honest Sharpe. Close with F-Squared: "this exact bug, undetected, was a \$35M fraud."

Risks. Full taint propagation through pandas is a rabbit hole; scope to a documented subset of operations and say so. Avoid resembling a generic backtester (the brief's named weak entry **BRIEF**) by never shipping strategy logic, only verification.

"Tapewatch": a feed health monitor demoed on a real disaster day

What it is. A streaming service that consumes a tick/quote feed and continuously scores its integrity: sequence gaps, stale quotes, crossed or locked books, price-band-violating prints, event-time vs receive-time skew, message-rate microbursts against declared capacity, and cross-feed divergence. A dashboard goes visibly red as a replayed historical incident unfolds, with per-anomaly drill-down. The design posture, per Databento's argument, is detect and flag, never silently clean [D14].

Evidence chain.

- The flash-crash report documents that data-integrity doubt itself withdrew liquidity [D1], therefore feed health is a first-class trading input, not ops hygiene.
- The SEC's NYSE order shows a 90-minute detection blind spot at the exchange itself [D5], therefore even the venue needs the tool you are demoing.
- Jane Street explains that gap handling is the subscriber's problem by design [D17], and OPRA microbursts now exceed most firms' receive capacity [D12], therefore gaps are certain, and detection is the tractable half.
- Great Expectations and pandera have no concept of sequence numbers or crossed books [D21], therefore the market-native check library is the gap.
- Free replay data exists: IEX full-day pcaps, LOBSTER order-book samples, Binance full-depth, Databento trial credit for 3 June 2024 [D22-D26], therefore the disaster-day demo is feasible without spend.

Build plan. Friday: choose the replay day (3 June 2024 for the Berkshire print, or a crypto flash-crash day for fully free depth data); write the feed parser. Saturday: the check engine (each check is a small invariant) and the scoring stream. Sunday: dashboard with a replay scrubber; drill-down that shows the exact bad prints; record the run.

Demo script. Start the replay on a calm morning, all green. Hit the incident: watch gap counts spike, the Berkshire print flagged inside a second, and the health score collapse, while narrating that the real exchange took 90 minutes to notice its own failure of this kind [D5].

Risks. Data wrangling can eat Saturday; pick one venue and three to five symbols. Timestamp discipline matters; keep event time and receive time as separate first-class columns from the first commit.

"Exchange-in-a-Box": a deterministic market-replay test harness with fault injection

What it is. A test harness that lets any trading algorithm be run against a simulated exchange that (a) replays real historical market data deterministically, (b) injects configurable faults (sequence gaps, duplicate and reordered packets, feed A/B divergence, stale snapshots, an exchange outage mid-session, a Berkshire-style bad print), and (c) produces bit-reproducible runs from a seed, so every failure is replayable. Ship with a scenario library of famous disasters and a scorecard: which faults did the algorithm under test survive?

Evidence chain.

- Jane Street: tests miss "the critical scenario that will expose your mistake," and trading systems are "big, complicated, weird state machines" [F8, F9]; HRT must simulate hardware before it trades [F11]; Optiver builds CI observability by hand [F12], therefore three elite firms confess the pain in their own voices.
- No off-the-shelf deterministic exchange simulator exists; the only generic entrant is domain-agnostic and enterprise-priced [F9], therefore the tooling landscape is empty at hackathon-reachable scale.
- Knight Capital: an untested deployment path cost \$460M in 45 minutes, and the SEC order names the missing checks [O1], therefore the scenario library opens with Knight.
- RTS 6 requires documented non-live algo testing and supervisors found firms' test environments wanting [O11, O12], therefore the harness output doubles as compliance evidence, which is your wedge for why small firms would adopt it.

Build plan. Friday: deterministic event-loop core (single-threaded, seeded, virtual clock) and a replay feed from LOBSTER or Binance data. Saturday: the fault injectors and a naive market-making bot as the test subject. Sunday: the scenario library (Knight-style runaway repeat, gap storm, bad print, outage), the scorecard, and the demo recording.

Demo script. Run the bot on clean replay: profitable, green. Same bot, same seed, inject the bad-print fault: watch it buy the fake dip and die. Show the run replayed bit-for-bit from the seed, then the scorecard across all scenarios. Close: "every firm builds this in-house; nothing like it exists off the shelf, and the SEC's Knight order is what the gap costs."

Risks. The most ambitious of the four; the mitigation is ruthless scope (one asset, one bot, five faults). Judges may not know the domain, so the Knight story must carry the first minute.

3.5 Alternates, briefly

Alert Linter. A static analyzer for Prometheus/Alertmanager rule sets that catches silently-failing patterns: absence-unaware rules, dead alerts referencing vanished metrics,

duplication drift. Evidence: HRT's own confession [P5] plus Knight's 97 ignored emails [O1]. Small, sharp, near-zero competition; the demo is running it on a real open-source rule corpus from GitHub and printing findings.

Algo Register. A git-integrated algo inventory and change-audit tool that drafts the MiFID II RTS 6 annual self-assessment article-by-article, gaps highlighted. Evidence: the FCA found firms cannot inventory their own algos [O10], the CBI found the self-assessments deficient [O11], ESMA calls the burden out [O12], and Two Sigma's \$255M shows what uncontrolled model change costs [O2]. Fully synthetic-data-friendly; no market data needed.

Calendar Consistency Engine. Cross-check exchange calendars against each other and against observed market data (volume on a day the calendar says closed?), emit diffs and alerts. Evidence: the endless correction changelog [P11]. The most under-the-radar pick in this dossier; trivially demoable.

Corporate-Actions Diff Engine. Detect when a vendor's adjusted price history silently changed, and flag which downstream backtests and positions are affected. Evidence: 4-in-10 straight-through rate and "large teams of personnel" [O18, P15]; restatement pain in vendor forums [P15]. Demo on two snapshots of a free EOD source around a split.

Bus-Factor Dashboard (knowledge risk). Ingest a synthetic firm's repo, notebooks and chat corpus; compute per-strategy knowledge concentration (who is the sole toucher of which components); simulate a departure and show what fraction of P&L-attributed code loses its last knowledgeable owner. Evidence: the Jane Street complaint's >50% profit drop after two departures [T1], churn rates [T8, T9], and the proof that structured capture works [T13, T14]. Pick this only if you accept sharing the brief-steered direction with other teams **BRIEF** and win on primary-source rigor.

Research Citations

The complete source list, grouped by code prefix. Each entry: code, type, source, the key claim it supports, URL. Entries marked "verify manually" were paywalled or bot-blocked during research and should be opened in a normal browser before quoting. Forum entries are colour-grade: use for illustration, not as load-bearing evidence.

R · Reproducibility and backtest overfitting

- R1** PAPER · Bailey, Borwein, Lopez de Prado, Zhu, "Pseudo-Mathematics and Financial Charlatanism," Notices of the AMS 61(5), 2014. Ten trials of a zero-skill strategy yield expected in-sample Sharpe 1.57; with 5 years of data, more than ~45 configurations near-guarantees illusory Sharpe 1. <https://www.ams.org/notices/201405/rnoti-p458.pdf>
- R2** PAPER · Bailey, Borwein, Lopez de Prado, Zhu, "The Probability of Backtest Overfitting," Journal of Computational Finance, 2015. A computable probability that a backtest is overfit (CSCV method). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2326253
- R3** PAPER · Bailey, Lopez de Prado, "The Deflated Sharpe Ratio," Journal of Portfolio Management 40(5), 2014. Corrects reported Sharpe for number of trials and non-normality; requires the trial count N that no workflow records. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2460551 · free PDF: <https://www.davidhbailey.com/dhbpapers/deflated-sharpe.pdf>
- R4** PAPER · Harvey, Liu, Zhu, "...and the Cross-Section of Expected Returns," Review of Financial Studies 29(1), 2016. Catalogues 316 factors; argues t-stat hurdle should be 3.0; "most claimed research findings in financial economics are likely false." https://people.duke.edu/~charvey/Research/Published_Papers/P118_and_the_cross.PDF
- R5** PAPER · Harvey, "Presidential Address: The Scientific Outlook in Financial Economics," Journal of Finance 72(4), 2017. p-hacking is endemic in finance research. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2893930
- R6** PAPER · Lopez de Prado, "The 7 Reasons Most Machine Learning Funds Fail" (2017) and "The 10 Reasons..." (JPM 2018). The Sisyphean paradigm: siloed researchers running thousands of undocumented backtests. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3031282 · https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3104816
- R7** PAPER (PLATFORM DATA) · Wiecki, Campbell, Lent, Stauth, "All that Glitters Is Not Gold," 2016. 888 real algorithms; in-sample Sharpe predicts out-of-sample with R-squared < 0.025; more backtests, bigger gap. <https://www.ssrn.com/abstract=2745220> · summary: <https://quantpedia.com/quantopians-academic-paper-about-in-vs-out-of-sample-performance-of-trading-alg/>
- R8** PAPER · Kapoor, Narayanan, "Leakage and the Reproducibility Crisis in ML-Based Science," Patterns, 2023. Taxonomy of data leakage across 300+ papers in 17 fields; verify exact counts before quoting. <https://arxiv.org/abs/2207.07048>
- R9** PAPER (PROTOCOL) · Arnott, Harvey, Markowitz, "A Backtesting Protocol in the Era of Machine Learning," Journal of Financial Data Science, 2019. The checklist practitioners have no tooling for. https://people.duke.edu/~charvey/Research/Published_Papers/P138_A_backtesting_protocol.pdf
- R10** PAPER · "How should you discount your backtest PnL?" arXiv 1902.01802. Practitioner haircut formulas for multiple testing. <https://arxiv.org/pdf/1902.01802.pdf>
- R11** BLOG + DEMO · Mathematical Investor (Bailey and Lopez de Prado), "How backtest overfitting in finance leads to false discoveries," 2022; interactive overfitting demo. <https://mathinvestor.org/2022/01/how-backtest-overfitting-in-finance-leads-to-false-discoveries/> · <https://carma.newcastle.edu.au/backtest/>
- R12** FORUM (COLOUR, BUT UNUSUALLY STRONG) · "I've reproduced 130 research papers about predicting the stock market" (2019) and its Hacker News discussion (701 points). "Literally every single paper was

either p-hacked, overfit, or a subsample of favourable data was selected." <https://news.ycombinator.com/item?id=20720095>

- R13** FIRM RESEARCH NOTE · Man Group with Campbell Harvey, "Overfitting and Its Impact on the Investor," 2015. A top systematic manager treating overfitting as an operating risk. <https://www.man.com/insights/overfitting-and-its-impact-on-the-investor>
- R14** CONFERENCE TALK · James Munro (Man Group), "Why a Hedge Fund Built Its Own Database," InfoQ. Data versioning built specifically so research can be reconstructed as-of research time. <https://www.infoq.com/presentations/arcticdb/>
- R15** PRACTITIONER BLOG · Andreas Clenow, "Backtesting: Is Zipline Dead?" 2021. The dominant open backtester orphaned by Quantopian's death; strategies unrunnable until a volunteer fork. <https://www.followingthetrend.com/2021/12/backtesting-is-zipline-dead-or-does-it-just-need-a-reload/>
- R16** PRACTITIONER SLIDES · Ernest Chan, "Backtesting and its Pitfalls." Survivorship demo: +388% biased vs -42% true universe; "using tomorrow prices to trade today." <https://epchan.com/img/links/Backtesting-and-its-Pitfalls.pdf>
- R17** PODCAST · Robert Carver (ex-Man AHL), Better System Trader ep. 026. "More complicated rules are more likely to have been overfitted." <https://bettersystemtrader.com/026-robert-carver/>
- R18** PLATFORM DOCS · QuantConnect Research Guide. A backtesting platform formally warning users about data dredging and hyper-tuning. <https://www.quantconnect.com/docs/v2/writing-algorithms/key-concepts/research-guide>
- R19** FORUM (COLOUR) · QuantConnect forum on survivorship bias creeping in through universe-selection APIs even on point-in-time data. <https://www.quantconnect.com/forum/discussion/13894/survivorship-bias-issues-in-universe-selection-equities/>
- R20** FORUM (ARCHIVED, COLOUR) · Quantopian community thread on Lopez de Prado's "7 Reasons"; the archive exists only because volunteers mirrored a dead platform. <https://quantopian-archive.netlify.app/forum/threads/the-7-reasons-most-machine-learning-funds-fail-by-marcos-lopez-de-prado.html>
- R21** NEWS · "How Quantopian died," Business of Business, 2021. Ten million community backtests never produced deployable alpha; pairs with R7 on why. <https://www.businessofbusiness.com/articles/how-quantopian-died-shut-down-quant-investment-robinhood/>
- R22** CODE REPOS · zipline (frozen upstream) and zipline-reloaded (community rescue fork). <https://github.com/quantopian/zipline> · <https://github.com/stefan-jansen/zipline-reloaded>
- R23** GITHUB ISSUES (PRIMARY) · Microsoft qlib cannot reproduce its own published benchmarks: issues #1668, #1782, #1890, #1383, all open or unresolved. <https://github.com/microsoft/qlib/issues/1668>
- R24** BLOG · "Did you know DVC doesn't handle large datasets?" (Meints). Generic MLOps versioning chokes on data-intensive research. <https://fizzylogic.nl/2023/01/13/did-you-know-dvc-doesn-t-handle-large-datasets-neither-did-we-and-here-s-how-we-fixed-it>
- R25** MISC (WEAK) · Evidence of MLflow/W&B friction at notebook scale (notebook-viewer shims; Show HN complaints). Use as supporting colour only; the stronger argument is the absence of any quant-aware tracker. <https://pypi.org/project/flown/> · <https://news.ycombinator.com/item?id=43851780>
- R26** CODE · Open-source PBO/DSR implementations, integrated into no backtester: pypbo; R package pbo. <https://github.com/esvhd/pypbo> · <https://cran.r-project.org/web/packages/pbo/readme/README.html>
- R27** SEC ACTION · AXA Rosenberg, 2011: \$242M settlement over a concealed quant model coding error; \$217M client losses. <https://www.sec.gov/news/press/2011/2011-37.htm>
- R28** SEC ACTION · F-Squared Investments, 2014: \$35M settlement; advertised track record was a backtest inflated by lookahead bias; firm went bankrupt. <https://www.sec.gov/newsroom/press-releases/2014-289>
- R29** CROSS-REFERENCE · Knight Capital deployment failure: see O1.
- R30** INDUSTRY STUDIES · Vanguard: indexes averaged +12.25% annual excess pre-launch, -0.26% post-launch; Morningstar study of 782 indexes found the same pattern. [https://tw.morningstar.com/tw/news/115894/on-back-tests-\(part-1\).aspx](https://tw.morningstar.com/tw/news/115894/on-back-tests-(part-1).aspx) · <https://www.etfstream.com/articles/index-outperformance-ends-when-funds-launch>

A · Alpha decay, crowding, and model drift

- A1** PAPER · McLean, Pontiff, "Does Academic Research Destroy Stock Return Predictability?" Journal of Finance 71(1), 2016. 97 predictors: 26% lower out-of-sample, 58% lower post-publication. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2156623
- A2** PAPER · Khandani, Lo, "What Happened to the Quants in August 2007?" Contrarian strategy returns eroded ~90% from 1995 to 2007; the coordinated unwind mechanism. http://web.mit.edu/~alo/www/Papers/august07b_2.pdf
- A3** PAPER · Hou, Xue, Zhang, "Replicating Anomalies," Review of Financial Studies, 2020. 65% of 452 anomalies fail replication under stricter tests; double-check the exact figure against the PDF before slides. <https://theinvestmentcapm.com/uploads/1/2/2/6/122679606/houxuezhang2019rfs.pdf>
- A4** PAPER · Jensen, Kelly, Pedersen, "Is There a Replication Crisis in Finance?" Journal of Finance, 2023. 82.4% replication rate, but post-publication alpha still falls roughly 47%. https://research-api.cbs.dk/ws/portalfiles/portal/95651880/theis_ingerslev_jensen_et_al_is_there_a_replication_crisis_in_finance_publishersversion.pdf
- A5** PAPER · Daniel, Moskowitz, "Momentum Crashes," Journal of Financial Economics, 2016. 2009: losers +163% vs winners +8%; regime-conditioning roughly doubles Sharpe. https://www.kentdaniel.net/papers/published/jfe_16.pdf
- A6** PAPER (PRACTITIONER) · Falck, Rej, Thesmar (CFM), "When Do Systematic Strategies Decay?" Quantitative Finance, 2022. Out-of-sample decay is predictable; higher in-sample Sharpe, larger decay. Publisher blocks bots; verify manually. <https://www.tandfonline.com/doi/full/10.1080/14697688.2022.2098810>
- A7** PAPER · Lou, Polk, "Comomentum: Inferring Arbitrage Activity from Return Correlations." Crowding measured from return co-movement; high crowding predicts low subsequent returns and negative skew. <https://personal.lse.ac.uk/polk/research/comomentum.pdf>
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- D24 DATA** · IEX HIST/DEEP: free full-day raw exchange pcaps back to 2016; open-source parsers exist. <https://iextrading.com/trading/market-data/>
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P · Plumbing and practitioner sweep

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- P4 FORUM (COLOUR)** · QuantNet, "Quant Developers (what is it like)": "40% of my time is spent just hunting down bugs." <https://quantnet.com/threads/quant-developers-what-is-it-like.3101/>
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O · Operational, risk, and regulatory

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- O8 SEC ACTION** · Athena Capital Research, 2014: first HFT manipulation case; the surveillance-of-your-own-algos problem. <https://www.sec.gov/newsroom/press-releases/2014-229>
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- O14** RULE + ANALYSIS · SEC Rule 15c3-5 adopting release (the pre-trade control requirements list); WilmerHale on its persistent ambiguity; FINRA exam findings. <https://www.sec.gov/files/rules/final/2010/34-63241.pdf> · <https://www.finra.org/rules-guidance/guidance/reports/2022-finras-examination-and-risk-monitoring-program/market-access-rule>
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- O17** INDUSTRY REPORT · SIFMA/ICI/DTCC T+1 After Action Report, 2024: same-day affirmation 73% to ~95%; NSCC clearing fund down \$3.0B. <https://www.sifma.org/news/press-releases/sifma-ici-and-dtcc-release-t1-after-action-report-industry-coordination-led-to-successful-transition-reducing-risk-and-costs-in-the-system>
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- O19** VENDOR (LABEL AS SUCH) · Gresham, Axoni, Duco on trade breaks and post-trade reconciliation. <https://www.greshamtech.com/blog/why-trades-fail-reconciliation>
- O20** INDUSTRY SURVEY · Acuiti panels: 92% of prop firms find reporting increasingly burdensome; >25% considered surrendering MiFID licenses; EU C-suite spends 3x US time on regulation. <https://www.acuiti.io/prop-firms-consider-giving-up-mifid-license/>
- O21** INDUSTRY SURVEY · Acuiti/Eventus Trade Surveillance Market Review: analysts spending 30+ hours/week manually closing alerts; 94% report rising complexity. <https://www.acuiti.io/acuiti-and-eventus-trade-surveillance/>
- O22** VENDOR/CONSULTANT (LABEL AS SUCH) · Solidus Labs buy-vs-build; Eventus and Kroll on RTS 6 practice: existing solutions are enterprise-priced and consultant-shaped. <https://www.soliduslabs.com/post/buy-vs-build-trade-surveillance>

T · Talent, churn, and knowledge transfer

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- T3** REGULATOR + COMMENTARY · SEBI interim order v. Jane Street, July 2025: ~\$4.3B India profits Jan 2023-Mar 2025; ~\$566M impounded. Epilogue on what the contested knowledge was worth. <https://blogs.law.ox.ac.uk/oblb/blog-post/2025/07/jane-street-and-expiry-day-trap-unpacking-sebis-crackdown-algorithmic>
- T4** NEWS · Bloomberg, Aug 13, 2026: Citadel imposes 1-2 year non-competes even on some analysts; escalation timeline. Paywalled; corroboration: <https://www.briefs.co/news/citadel-s-two-year-ban-on-job-moves-stalls-analyst-careers/>

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- T9 TRADE PRESS (WSJ-DERIVED)** · Hedgeweek, 2025: nine-figure packages; PMs cut at 7.5% drawdown. <https://www.hedgeweek.com/multi-manager-hedge-funds-escalate-talent-war-arms-race-with-nine-figure-pay-packages/>
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- T13 COMPANY PRIMARY** · WorldQuant BRAIN and the Global Alphathon: alpha research made machine-legible by construction; 650+ consultants, 34,000+ data fields. <https://www.businesswire.com/news/home/20220801005832/en/WorldQuant-Launches-BRAIN-Platform-and-Inaugural-Global-Alphathon-Competition> · <https://www.worldquant.com/brain/>
- T14 INDUSTRY ANALYSIS** · Rubinstein, "Alpha Capture," Net Interest: Marshall Wace TOPS, ~500,000 structured ideas/year by 2005, ~\$30B AUM; structured capture of trading intuition works. <https://www.netinterest.co/p/alpha-capture>
- T15 CROSS-REFERENCE** · Man Group research data sharing (ArcticDB): see F20. Firms invest heavily in data sharing while nothing equivalent exists for reasoning.
- T16 ACADEMIC MONOGRAPH** · MacKenzie, "Trading at the Speed of Light" (Princeton, 2021), built on 300+ insider interviews: HFT knowledge is tacit and socially embedded. Your interviews-by-proxy. <https://press.princeton.edu/books/hardcover/9780691211381/trading-at-the-speed-of-light>
- T17 FORUM (COLOUR ONLY)** · Blind and WSO threads on non-compete enforcement, onboarding pain, Millennium pod tenure. Anonymous; never load-bearing. <https://www.wallstreetoasis.com/forum/hedge-fund/high-turnover-at-millennium>
- T18 VENDOR SURVEY (CAVEAT)** · Panopto/YouGov: inefficient knowledge sharing costs large firms \$47M/year; generic corporate, not quant-specific. <https://www.panopto.com/company/news/inefficient-knowledge-sharing-costs-large-businesses-47-million-per-year/>
- T19 NEWS** · Fortune, 2023: Citadel interns at \$19,200/month; 69,000 applicants, under 1% acceptance. <https://fortune.com/2023/08/28/citadel-internship/>
- T20 ECOSYSTEM EVIDENCE** · The quant interview-prep arms race: QuantGuide and at least four competitors exist solely to drill firm-tagged question banks. <https://www.quantguide.io/>
- T21 BLOG (WEAK)** · Practitioner criticism of quant interviews (speed brainteasers, false negatives). Thinnest leg; colour only. <https://paulieswildlife.medium.com/5-problems-with-quant-interviews-and-how-to-correct-them-cb4727cc25c5>
- T22 MARKET FACTS** · Glean at \$7.2B valuation / \$200M ARR: enterprise search indexes documents that exist; quant knowledge mostly is not written down. https://en.wikipedia.org/wiki/Glean_Technologies
- T23 TRADE PRESS** · HR literature on why wikis and exit interviews fail to capture tacit knowledge. <https://hrexecutive.com/unshared-knowledge-destroys-a-companys-bottom-line/>

T24 LAW-FIRM ALERT · Morrison Foerster on the Two Sigma order: the firm knew of model vulnerabilities for years; no systematic record of who changed which parameter when. <https://www.mofo.com/resources/insights/250226-top-5-sec-enforcement-developments>

F · Firm-published technical material

All items below are authored or hosted by the firms themselves. Signals and Threads (S&T) is Jane Street's podcast; full transcripts are public on each episode page. Millennium publishes no technical material; WorldQuant (its affiliate) is the closest primary channel (F25, F26). Tower Research's current public technical content is ML-community material rather than systems write-ups.

- F1** JANE STREET · PODCAST · "Multicast and the Markets" (S&T ep. 3): exchange feeds are UDP multicast; gaps, reordering and retransmission are the subscriber's problem by design. <https://signalsandthreads.com/multicast-and-the-markets/>
- F2** OPTIVER · TALK · Carl Cook, "When a Microsecond Is an Eternity" (CppCon 2017): the OS and hardware stack fights latency; profiling improvements guarantee nothing. <https://www.youtube.com/watch?v=NH1Tta7purM>
- F3** OPTIVER · TALK · David Gross, low-latency systems engineering: only measurement works. https://david-grs.github.io/low_latency_presentation/
- F4** IMC · BLOG · "How are FPGAs used in trading?": data volume and nanosecond timescales outrun software. <https://www.imc.com/us/articles/how-are-fpgas-used-in-trading>
- F5** HRT · BLOG · "Low Latency Optimization: Understanding Huge Pages" (parts 1-2): Linux memory management as a latency source. <https://www.hudsonrivertrading.com/hrtbeat/low-latency-optimization-part-1/>
- F6** JANE STREET · PODCAST · "Programmable Hardware" (S&T ep. 1): when software cannot keep up, hardware tooling becomes the bottleneck. <https://signalsandthreads.com/programmable-hardware/>
- F7** JANE STREET · PODCAST · "Performance Engineering on Hard Mode" (S&T ep. 18): bursty, latency-critical workloads break standard optimization playbooks. <https://signalsandthreads.com/performance-engineering-on-hard-mode/>
- F8** JANE STREET · BLOG · "Getting from tested to battle-tested": "It's all too easy for your tests to miss the critical scenario that will expose your mistake." <https://blog.janestreet.com/getting-from-tested-to-battle-tested/>
- F9** JANE STREET · PODCAST · "Why Testing is Hard and How to Fix it" (S&T ep. 26, with Antithesis): with threads and sockets "your computer program is not a pure function... It's a big, complicated, weird state machine." <https://signalsandthreads.com/why-testing-is-hard-and-how-to-fix-it/>
- F10** JANE STREET · BLOG · "The Joy of Expect Tests": test-writing friction leaves critical systems undertested. <https://blog.janestreet.com/the-joy-of-expect-tests/>
- F11** HRT · BLOG · "How We Verify Custom Hardware": FPGA trading logic must be simulated before it touches the market. <https://www.hudsonrivertrading.com/hrtbeat/verify-custom-hardware/>
- F12** OPTIVER · BLOG · "Building visibility into a CI platform": trading-code CI lacks the observability it needs out of the box. <https://www.optiver.com/insights/technology-blog/building-visibility-into-a-ci-platform/>
- F13** HRT · BLOG · "HeraclesQL": thousands of alerts; old rules "rarely handled timeseries absence correctly." <https://www.hudsonrivertrading.com/hrtbeat/heraclesql-a-python-dsl-for-writing-alerts/>
- F14** HRT · BLOG · "Scaling Prometheus to Billions of Samples a Second." <https://www.hudsonrivertrading.com/hrtbeat/scaling-prometheus/>
- F15** JANE STREET · BLOG · "Magic-trace": sampling profilers "approximately useless" for microsecond latency spikes. <https://blog.janestreet.com/magic-trace/>
- F16** JANE STREET · PODCAST · "Solving Puzzles in Production" (S&T ep. 20): incident response as cross-system detective work. <https://signalsandthreads.com/solving-puzzles-in-production/>

- F17** JANE STREET · BLOG · "Finding memory leaks with Memtrace": production-only pathologies in long-running trading services. <https://blog.janestreet.com/finding-memory-leaks-with-memtrace/>
- F18** HRT · BLOG · "Building a Distributed Filesystem for Scalable Research": hundreds of petabytes; vendor appliances too expensive. <https://www.hudsonrivertrading.com/hrtbeat/distributed-filesystem-for-scalable-research/>
- F19** TWO SIGMA · ARTICLE · "Treating Data as Code": dataset onboarding and versioning bottleneck research. <https://www.twosigma.com/articles/treating-data-as-code-at-two-sigma/>
- F20** MAN GROUP · BLOG · "Our Journey Creating ArcticDB": no off-the-shelf store for versioned research DataFrames existed, so a hedge fund built a database. <https://arcticdb.io/blog/our-journey-creating-arcticdb-solving-the-challenge-of-dataframes-at-scale/>
- F21** JANE STREET · PODCAST · "Building a Data Warehouse from Scratch" (S&T ep. 28): analytical queries against data arriving at trading speed. <https://signalsandthreads.com/building-a-data-warehouse-from-scratch/>
- F22** G-RESEARCH · OSS · Armada: research batch compute broke single-cluster Kubernetes; now a CNCF project used in production. <https://armadaproject.io/>
- F23** OPTIVER · BLOG · "Designing for latency and research iteration": the research-vs-production tension in the firm's own words. <https://www.optiver.com/insights/technology-blog/designing-for-latency-and-iteration/>
- F24** XTX · CORPORATE · Disclosed research infrastructure: 25,000+ GPUs, over 1 exabyte of storage. <https://www.xtxmarkets.com/>
- F25** WORLDQUANT · PAPER · Kakushadze, "101 Formulaic Alphas" (arXiv 1601.00991): industrialized alpha production, formulas as code. <https://arxiv.org/abs/1601.00991>
- F26** WORLDQUANT · PLATFORM · BRAIN: crowdsourced, machine-legible alpha research; the backtesting platform as the moat. <https://www.worldquant.com/brain/>
- F27** OPTIVER · BLOG · "Pushing Postgres beyond storage": propagating reference-data changes to thousands of consumers. <https://www.optiver.com/insights/technology-blog/pushing-postgres-beyond-storage/>
- F28** JANE STREET · PODCAST · "Clock Synchronization" (S&T ep. 6): auditable sync within 100 microseconds of UTC. <https://signalsandthreads.com/clock-synchronization/>
- F29** JANE STREET · PODCAST · "State Machine Replication, and Why You Should Care" (S&T ep. 14): determinism as the path to reliability. <https://signalsandthreads.com/state-machine-replication-and-why-you-should-care/>
- F30** JANE STREET · PODCAST · "Swapping the Engine out of a Moving Race Car" (S&T ep. 15): modernizing a system the firm cannot trade without; monthly rolls considered fast. <https://signalsandthreads.com/swapping-the-engine-out-of-a-moving-race-car/>
- F31** HRT · BLOG · "How Our Engineers Hot-Patched a Third Party Binary Library": vendor dependency in the critical path. <https://www.hudsonrivertrading.com/hrtbeat/how-our-engineers-hot-patched-a-third-party-binary-library/>
- F32** JANE STREET · BLOG · "How we accidentally built a better build system for OCaml": a monorepo growing 4M to 65M lines outran the build system. <https://blog.janestreet.com/how-we-accidentally-built-a-better-build-system-for-ocaml-index/>
- F33** HRT · BLOG · "Inside HRT's Python Fork": imports so slow they forked CPython. <https://www.hudsonrivertrading.com/hrtbeat/inside-hrts-python-fork/>
- F34** JANE STREET · PODCAST · "Building Tools for Traders" (S&T ep. 23): tooling for microsecond-domain expert users. <https://signalsandthreads.com/building-tools-for-traders/>
- F35** JANE STREET · PODCAST · "From the Lab to the Trading Floor" (S&T ep. 19): experts route around bad tools; knowledge stays in heads. <https://signalsandthreads.com/from-the-lab-to-the-trading-floor/>
- F36** JANE STREET · PODCAST · "Writing, Technically" (S&T ep. 10): internal documentation never keeps up. <https://signalsandthreads.com/writing-technically/>

Standing Context (for any future reader or agent)

This appendix makes the dossier fully self-contained: what the assignment is, what constraints apply, what has been decided, and what remains to be done. Someone (or some agent) picking this up cold should be able to proceed from here alone.

A.1 The assignment

The team is entering a university hackathon track called Fundamentum, themed on quantitative trading firms **BRIEF**. Per the track brief, the task is: pick ONE foundational problem quant firms actually live with, investigate it deeply using real sources, and build a weekend prototype that only makes sense because of the research. Required submission: (1) a working prototype grounded in the findings, (2) a research trail of actual sources, notes and records (evidence of the journey, not a tidy summary), (3) a video pitch walking through the problem, what the research uncovered, and the decisions that came from it. The on-theme prize goes to the clearest traceable line from foundational research to final build. The brief explicitly warns off two directions: latency ("make trades faster") and alpha models ("find a better signal"), as unbuildable and unresearchable in a weekend; it names as a weak entry "a generic backtesting toy or a fake trading dashboard, justified by intuition" **BRIEF**.

A.2 Team constraints and choices made so far

- The team has no industry contacts and will do **no interviews**. All evidence must be public and citable. The chosen substitutes, in order of weight: firm-published technical material, regulatory and court records, peer-reviewed papers, industry reports; forums only as colour.
- The team specifically asked for primary technical material from Jane Street, IMC, Optiver, Millennium and Tower Research (Part 2.8 and the F sources). Corporate financial disclosures were explicitly ruled out of scope by the team.
- Fact of record: Millennium publishes essentially no technical material; WorldQuant is its closest public research channel. Tower Research's public technical output is thin.
- No project has been selected yet. This dossier's ranking (Part 3) is a recommendation, not a decision.

A.3 Glossary of terms used in this dossier

TERM	MEANING
Sharpe ratio / IC	Risk-adjusted return measure / information coefficient, the correlation between a signal and subsequent returns. Both are the standard health metrics of a strategy.
Backtest, in-sample, out-of-sample	Historical simulation of a strategy; performance on the data used to build it vs on unseen data.

Lookahead bias / survivorship bias	Using information not yet available at simulated decision time / testing only on instruments that survived to the present. The two classic silent backtest killers.
Alpha, alpha decay, crowding	Excess return from a signal; its erosion over time; erosion specifically caused by many players exploiting the same signal.
SIP / direct feeds / NBBO	The consolidated US equities data feed (Securities Information Processor); exchanges' faster proprietary feeds; the national best bid and offer derived from quotes.
Sequence gap / A-B arbitration	Missing packets in a market data stream, detected via sequence numbers; reconciling the two redundant copies (A and B) of a feed.
LULD price bands	Limit-up-limit-down bands outside which prints are erroneous; the mechanism behind the June 2024 Berkshire incident (D6).
SIP entities: CTA/UTP; OPRA	The consolidated tape operators for US equities; the (enormous) consolidated options feed.
Rule 15c3-5 / RTS 6 / SR 11-7 / CAT	SEC pre-trade risk-control rule; MiFID II algo testing and self-assessment rules; Fed model-risk guidance; the Consolidated Audit Trail order-event reporting regime.
Pod shop / multi-manager	Hedge fund structured as many autonomous teams (Millennium, Citadel, Balyasny); the high-churn end of the industry.
Garden leave / non-compete	Paid idle period and post-employment restriction that firms use to slow knowledge leaving with people.
DSR / PBO / CUSUM / BOCPD	Deflated Sharpe Ratio; probability of backtest overfitting; two changepoint-detection methods (classical and Bayesian) usable for decay monitoring.
kdb+/q	The dominant commercial tick database and its language; expensive and locked-in (P13).

A.4 What a follow-on agent or teammate should do next

1. **Pick the project** (Part 3). The decision inputs are the three ratings per problem (Part 1.2) plus team taste. Nothing else blocks starting.
2. **Pull the primaries for the chosen project only.** Each project's evidence chain lists its sources; download the PDFs (SEC orders, papers) and screenshot the firm posts. Items marked "verify manually" are bot-blocked but open in a normal browser.
3. **Re-verify before slides:** the specific figures flagged in Parts 2 and 4 (A3's 65%-of-452, A10's capacity numbers, R8's counts), and any 2025-2026 items in case of later developments.
4. **Start the decision log immediately** (format in Part 1.5): it is a required deliverable in spirit and the thing the on-theme prize scores.
5. **Data setup per project:** Project 1 needs one CSV from openassetpricing.com (A26). Project 2 needs no external data. Project 3 needs one replay day (D22-D26). Project 4 needs LOBSTER or Binance depth data (D23, D25).
6. **Known gaps in this dossier:** no interviews (by constraint); Reddit-based practitioner threads were not collected (the team judged them low-value); AQR essays, Bloomberg articles and two paywalled papers were cited from corroborating coverage rather than

full text; the SEBI order (T3) was cited via commentary, and the primary PDF is retrievable from sebi.gov.in if needed.